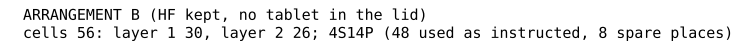


MESHESAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



CELL Samsung INR18650-35E: d 18.55 max, L 65.25 max, 50 g max
(spec Ver. 1.1. 3.10 and drawing)

BLOCK A06's construction (the ruled base block): cells touching, wrap 0.5 a side, joints 1.0 at each cell end along X (nickel strip and fish paper), layer 2 nested in the grooves

HOLD the block bonded to the lid plate (cell glue and a fillet, adhesive TBD); the cover on 8 M3 standoffs in the end bays presses a PORON 4701-30 pad 1.0; the plate bonded to the lid's ceiling with DP8005 as the r2 set (no hole in the case)

VENT every cell end faces its 1.0 joint; the end bays are open slots; the case relieves through Peli's pressure valve

STACK FROM THE CEILING	nominal
DP8005 bond	0.20
5052-H32 lid plate	2.00
block, one layer / two layers	19.55 / 35.61
PORON pad (compressed)	1.00
cover 1.0 + insulation 0.25	1.25
module depth, one / two layers	24.00 / 40.06
P2 on 3.0 standoff (ASSUMPTION)	22.97

ROOM face top to ceiling: 47.92 nominal, 44.39 worst,
42.11 sensitivity (frame seat.out, r2 record B)

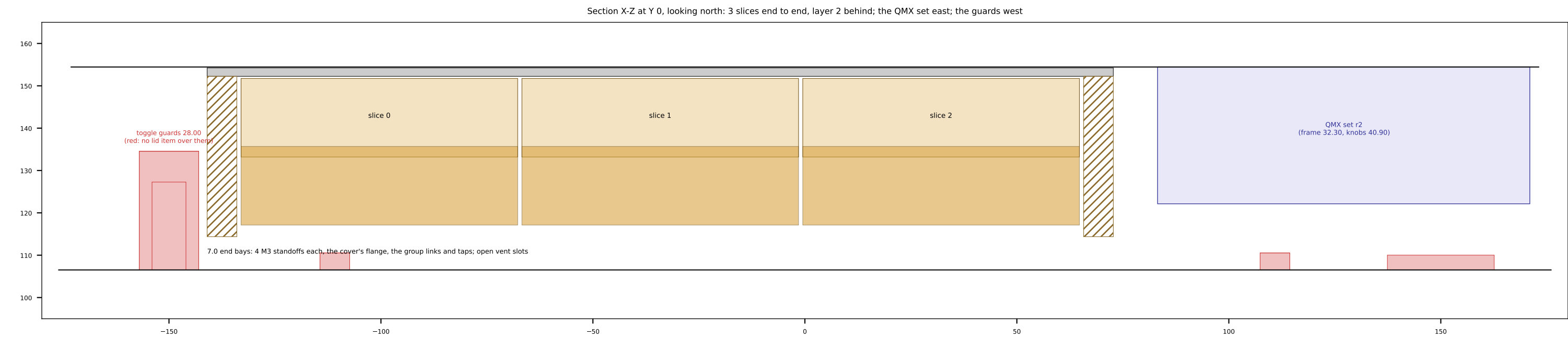
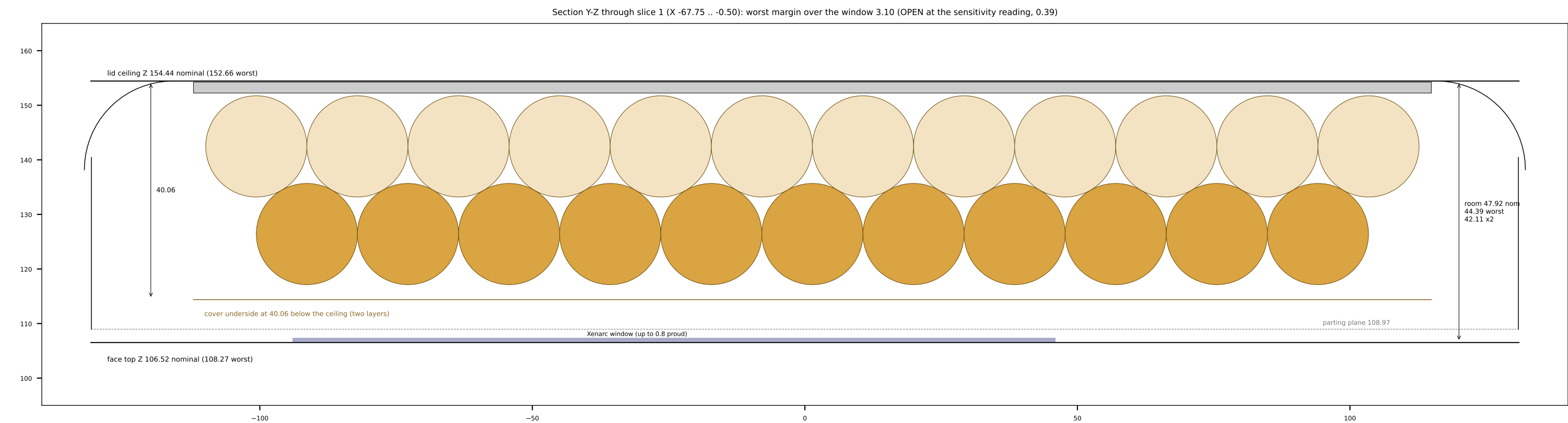
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TIGTEST ROWS (worst, x2; min 1.0)
two layers over the Xenarc window 0.80: 3.10, 0.39 OPEN
two layers over the LED bar 1.50: 2.40, -0.31 OPEN
two layers over the e-paper lens: 3.85, 1.14 MET
one layer over the XFRAME screws: 15.96, 13.25 MET
layer 2 refused over XFRAME 1 and 3 (4.00): -0.10 worst
```

PLAN module to the QMX tray 10.45; inside the flat ceiling
1.00 in Y (the fillets begin at +/-115.93)
west edge 2.0 from the toggle guard caps (X -143.0)
west slice stops north of the headset jacks (TBD)

OPEN AT THE MOCK-UP: T9 (chalk on the module's cover and the face parts, lid closed), T8 (pull test of a bonded plate on Peli's polypropylene), T-A1-1 to T-A1-4 of the README.

MeshSat field kit V2, Option A(i) lid pack: sheet A1-2 sections: the module under the ceiling over the face, at the worst room

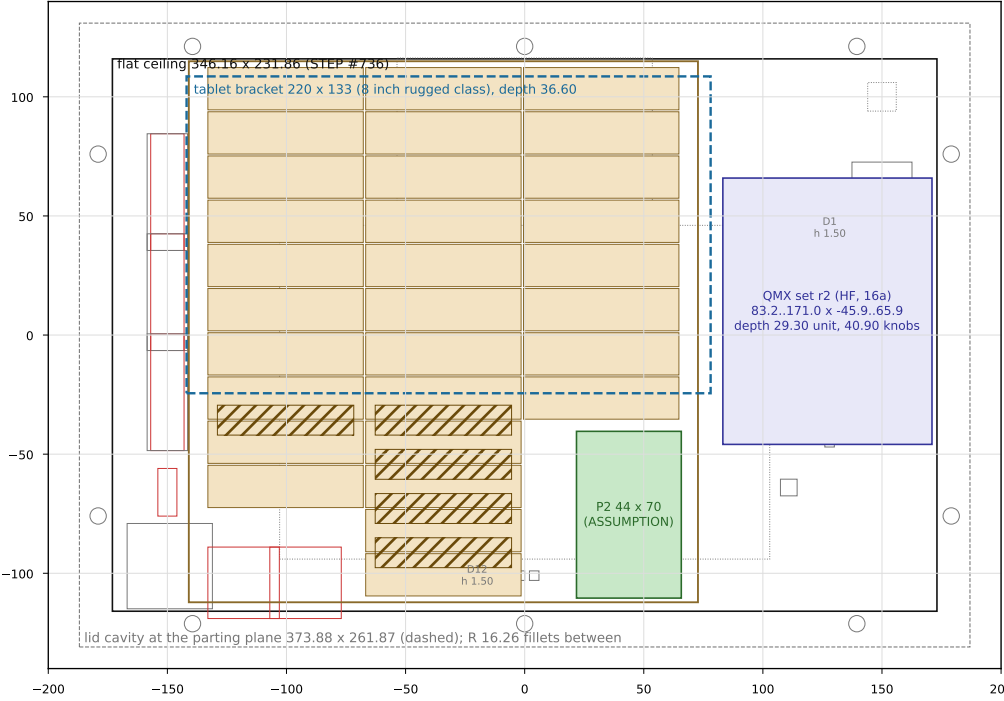
MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



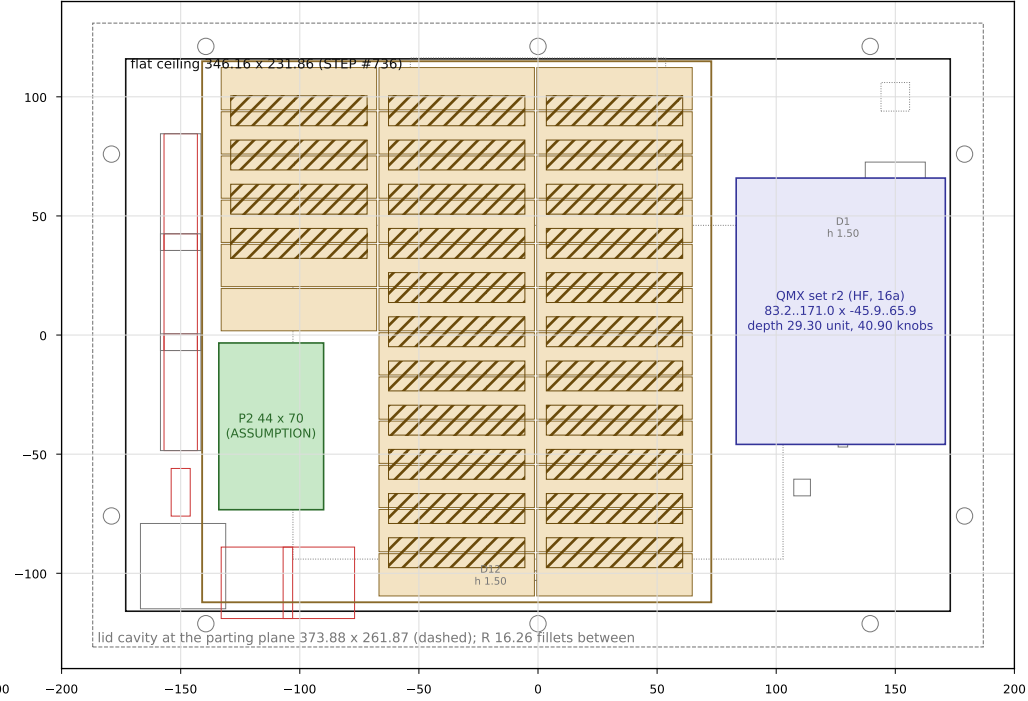
MeshSat field kit V2, Option A(i) lid pack: sheet A1-3 the arrangements that fit, and the owner decision

MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.

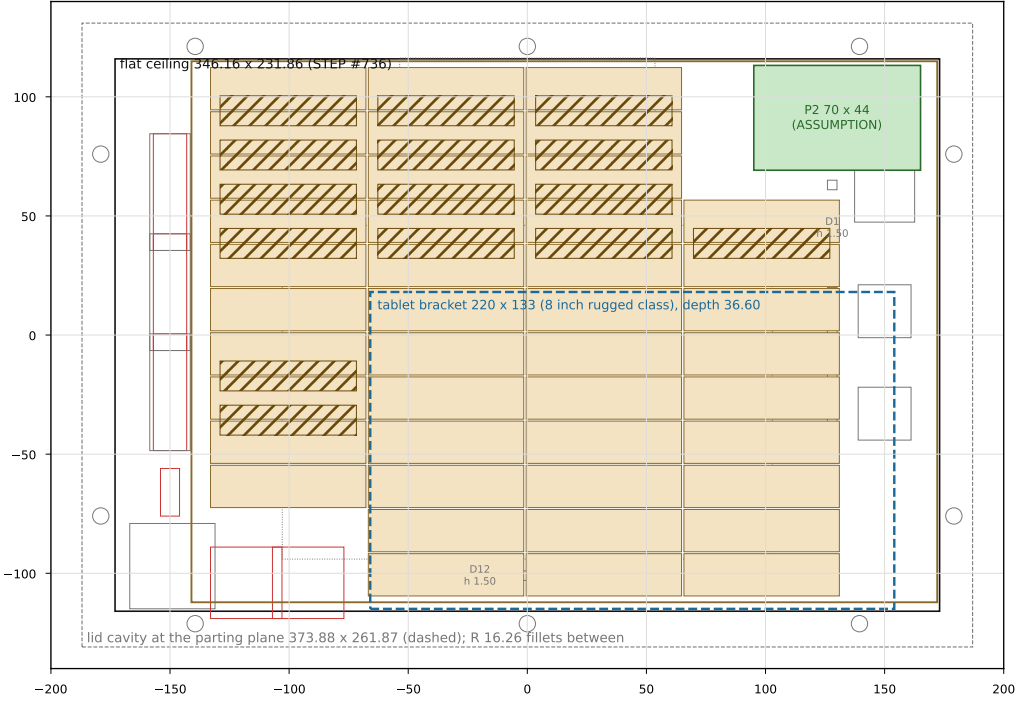
A: HF kept, 8 inch tablet kept
35 cells: 4S8P in the lid, 4S14P with the base



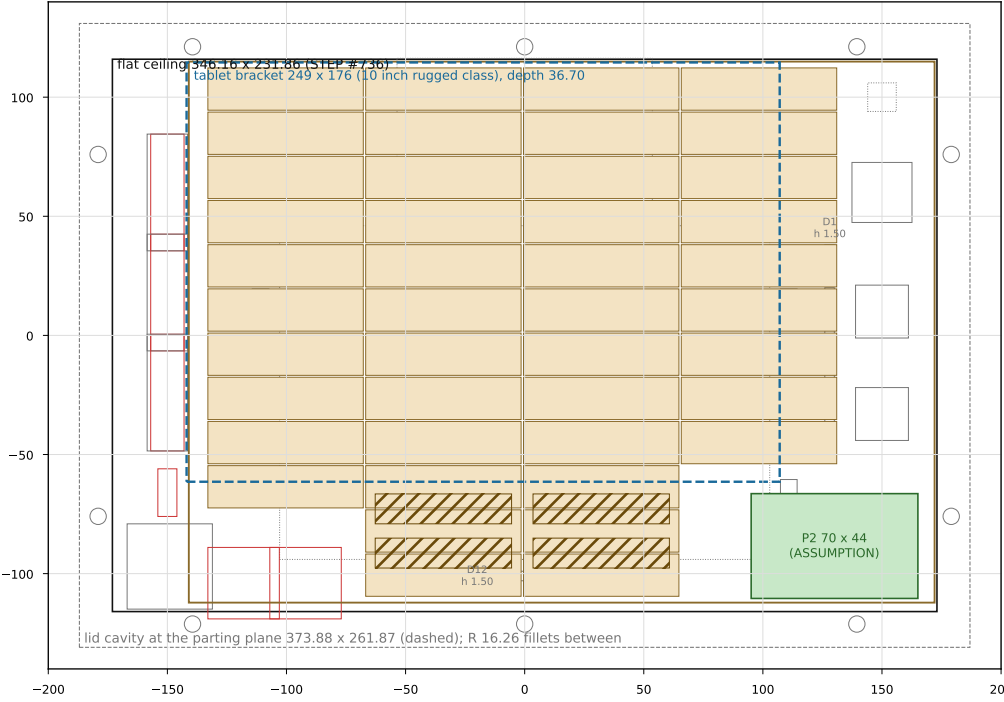
B: HF kept, no tablet in the lid
56 cells: 4S14P in the lid, 4S20P with the base



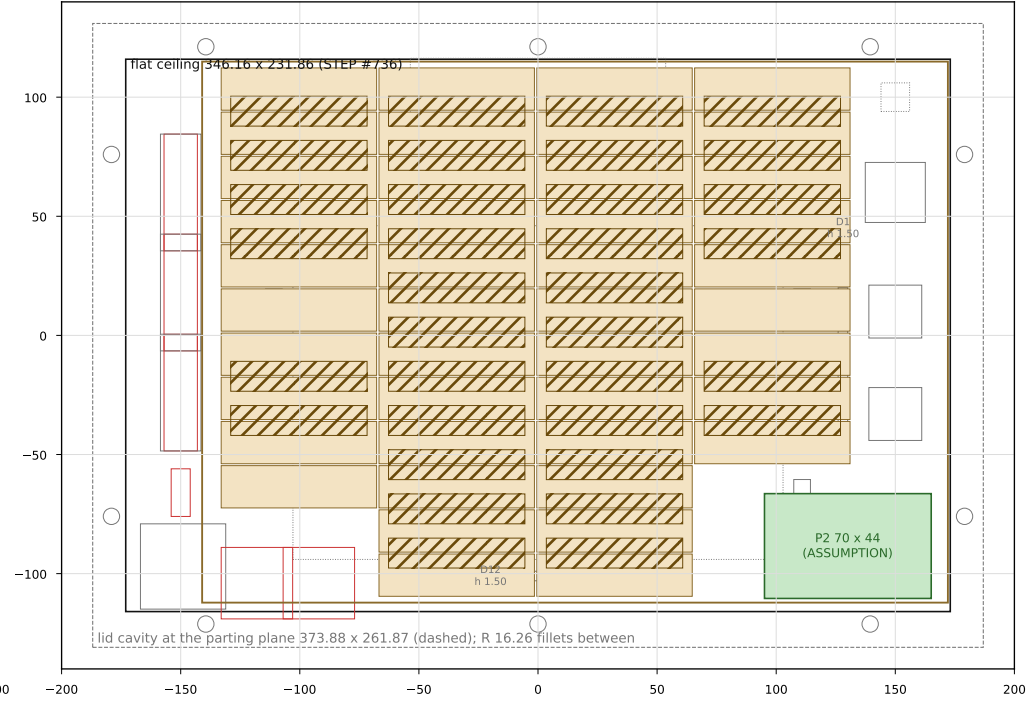
C: 8 inch tablet kept, HF out of the lid
58 cells: 4S14P in the lid, 4S20P with the base



C10: 10 inch tablet kept, HF out of the lid
47 cells: 4S11P in the lid, 4S17P with the base



D: HF and tablet both out of the lid
77 cells: 4S19P in the lid, 4S25P with the base



THE OWNER DECISION (the session does not take it:
it removes an owner-approved function, 6 Sep 2026)

48 cells (4S12P) in the lid with BOTH the QMX HF set (16a) and the tablet bracket (16d): DOES NOT FIT. At the worst room the lid holds 35 places with both, 4S8P; the total 4S14P is under section 9c's 4S15P floor.

What fits 4S12P (each with every row of sheet A1-1 at or above its minimum at the worst):
1. HF kept, tablet bracket out of the lid: 56 places (4S14P); the tablet goes out of the case.
2. 8 inch tablet kept, HF (QMX) out of the lid: 58 places (4S14P); HF inside (16a) is lost: the base has no volume for the QMX set (section 8)
3. Both kept, lid pack 4S8P: not Option A(i); M1 not met on section 9's model

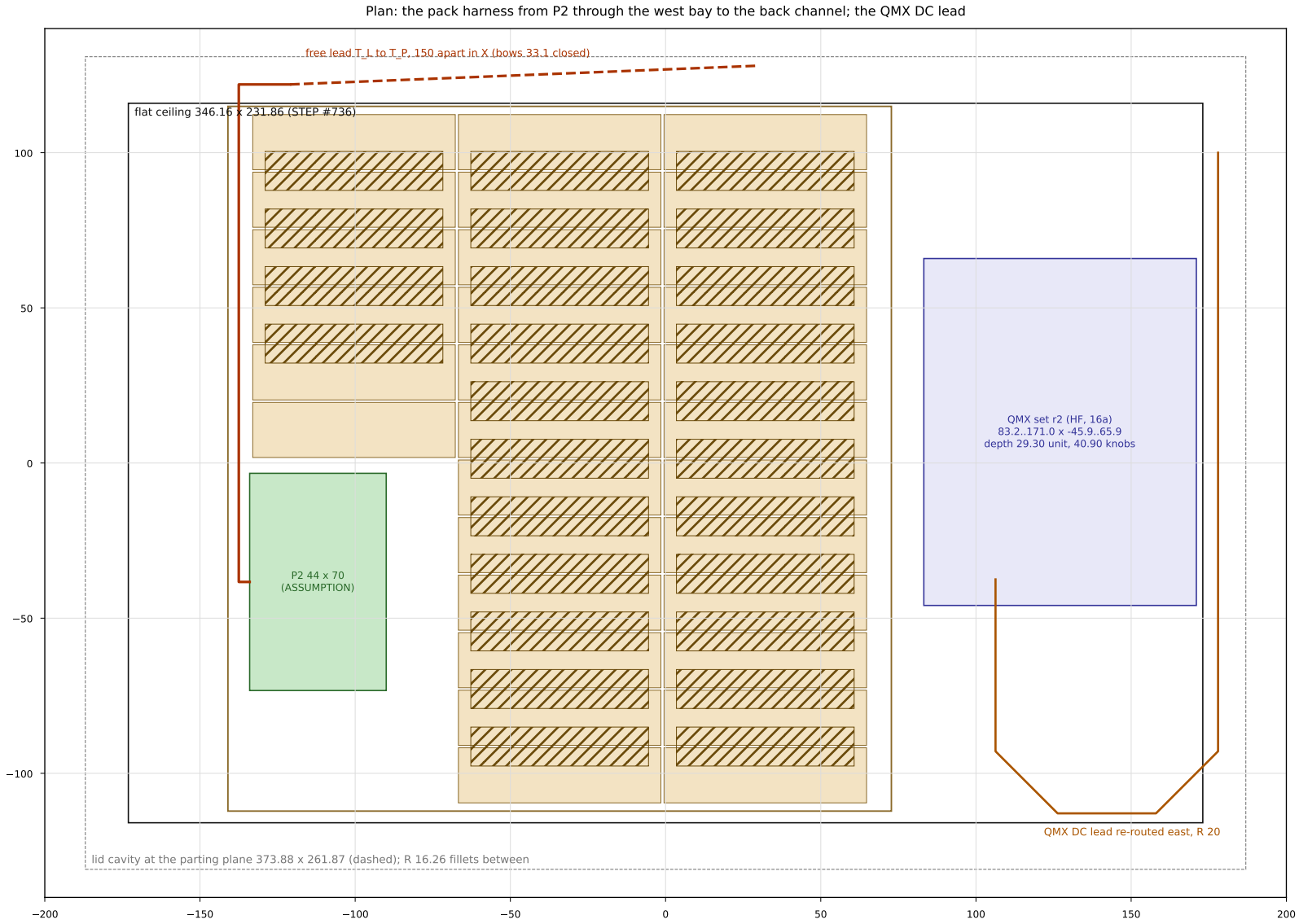
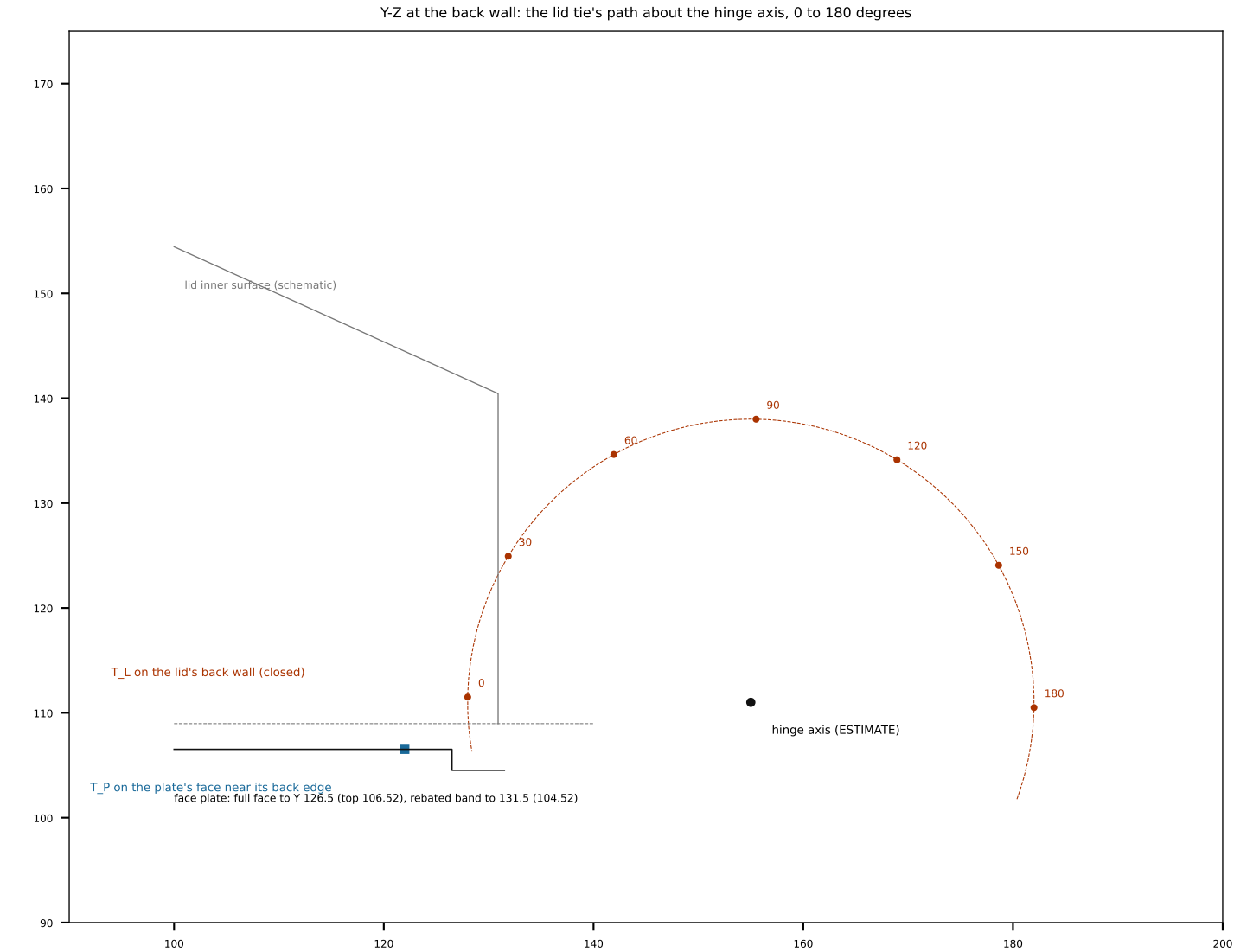
10 inch tablet: fits only with HF out: 47 places (4S11P). Both out: 77 places (4S19P).

Session's recommendation (engineering view only): option 1, which keeps a radio bearer and gives 8 spare places; the tablet works from outside the case on the kit's WiFi and USB-C outlet.

D-01 already defers HF, the tablet bracket and a second pack from prototype 1: Option A(i) reopens D-01 whichever option is taken.

MeshSat field kit V2, Option A(i) lid pack: sheet A1-4 the hinge harness: route, travel, ties, disconnect; the QMX DC lead re-routed

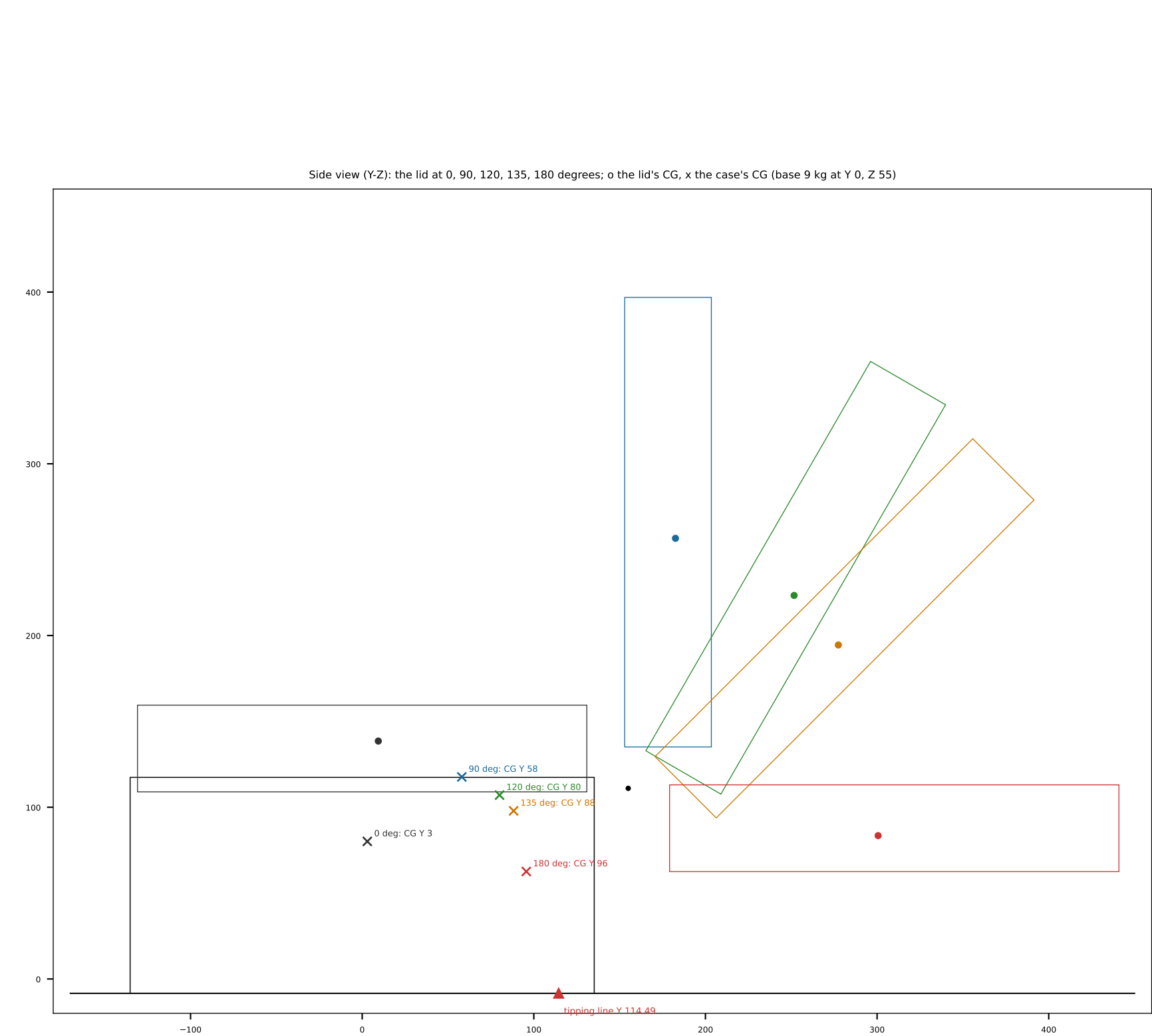
MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



CURRENT (ASSUMPTION for fnd/alelec): 10 A continuous, 18 A for 60 s, 8.6 A charge
CONDUCTOR 12 AWG fine-strand silicone, OD 4.4 (class); 2 x 0.80 m loop 8.34 mOhm:
150 mV and 2.7 W at 18 A; adiabatic rise 8.9 K over 60 s
TRAVEL the lid tie turns at R 27.0 about the axis; the chord T_L..T_P runs
150.2 closed to 161.6 at 180 degrees; lead 169.7 between the ties
BEND closed bow 33.1 (radius about 102), open S radius about 94,
against 10 x OD = 44 (the requirement on the pick)
STRAIN RELIEF bonded tie mounts at T_L, T_P and every 60 mm or less;
a printed guide keeps the lead inboard of the lid wall (Y < 130.9)
DISCONNECT XT60 pair and JST XH 1x4 beside P2 (the lid comes off)
CROSSING the sealed face: S-95 (EQ-31), NOT designed: +2 power poles
of 25 A or more and 4 signal contacts, sealed mated and unmated

MeshSat field kit V2, Option A(i) lid pack: sheet A1-5 mass and open-lid stability

MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



LID, arrangement B: 4.90 kg in all			
lid pack cells, 56 x 50 g max (spec 3.10)			2.800
lid plate 5052-H32 2.0 (2.68 g/cm3)			0.260
cover, printed V-0 class 1.0 (1.25 g/cm3, ESTIMATE)			0.105
nickel strip and group links (ESTIMATE)			0.097
wrap, fish paper, cell glue, pad, standoffs, bond (E			0.114
P2 protection board (ASSUMPTION, board P class)			0.080
lid harness to the hinge: 2 x 12 AWG 0.45 m, sense l			0.080
QMX set r2: unit and plugs 0.240, tray, frame, plate			0.372
lid shell, 39.7 percent of 2.5 kg (ESTIMATE)			0.993

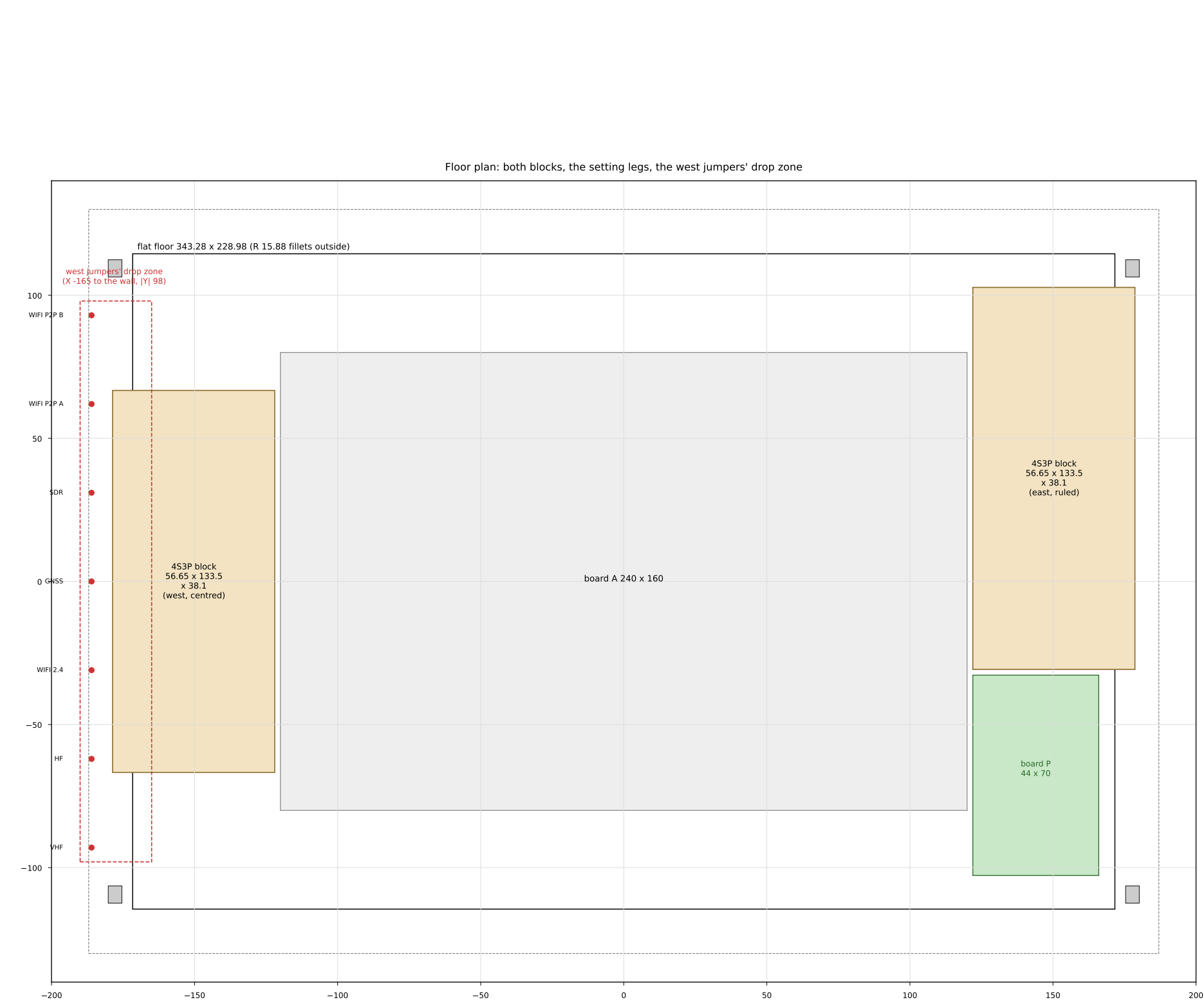
WORST OVER THE SWEEP (base 7..14 kg, CG Y -20..+20, Z 45..65, hinge axis Y 147..163): critical back slope, deg			
lid open	B	A	
90	14.9	17.1	
100	11.6	14.0	
110	8.4	11.1	
120	5.4	8.4	
135	1.3	4.9	
150	-2.2	2.2	(tips on level ground)
180	-5.0	1.7	(tips on level ground)

(the hinge axis at Y 155 here; the .out's table also sweeps 147..163)

VERDICT: stands on level ground to 120 degrees of opening in every swept case; a light base tips from 135. FIX (SESSION, reversible): a lid stay stopping the lid at 100 degrees (webbing strap between two bonded anchors, lid back wall and plate back band; no hole; about EUR 10, ESTIMATE). With it the open case stands on a slope of about 10 degrees at the worst (lid_pack_a1.out section 4). The stop angle, the hinge axis and the case's CG are read on the mock-up (T-A1-3); Peli states no load for its hinges (T-A1-4).

MeshSat field kit V2, Option A(i) lid pack: sheet A1-6 the base pockets: section 8's 4S6P at the same worst figures

MESH SAT-1357, stream a1mech, 29 Sep 2026. Case frame: X east, Y toward the hinge (back) wall, Z up from the base floor, lid closed. mm. Prototype design, AI review: nothing built, bought or fitted; analytical success is not physical verification.



ROW	worst	x2	verdict
M4a	1.85	0.85	OPEN: placed by hand; T4
M4aw	1.85	0.85	OPEN: placed by hand; T4 (mirror: Peli's cavity and
M4b	7.38	6.38	MET
M4bw	7.38	6.38	MET (mirror: Peli's cavity and the RF entry plates a
M4c	3.18	1.80	MET
M4cw	3.18	1.80	MET (mirror: Peli's cavity and the RF entry plates a
M5	1.77	0.27	OPEN: placed by hand, the legs' locator; T4
M5w	37.77	36.89	MET (the block's Y place is not ruled; centred here)
M6	3.66	2.90	MET
M6w	3.99	n/a	OPEN: the x2 reading is not computed by A06's tool;

WEST RF JUMPERS: 5 of 7 west cables (Y -62, -31, 0, 31, 62) fall onto the west block; the slot over it is 7.19 at the worst less C33's zone (3.99); the only drop east of it is its 2.0 gap to board A, under one RG-316 (2.49): FAILS AS ASSUMED, held OPEN (section 8b, the M17 class): the west RF entry is re-planned before the west block is taken. Hold-down S-27 open for both blocks.

The mirror rows hold only because Peli's cavity is the same at both ends (ribs at Y 0 and +-76.2 on both end walls, R 15.88 on all four floor edges: VERIFIED) and the RF entry plates share one outline (panel1450 C4).